E-COMMERCE PROFIT DIAGNOSTIC · SAMPLE REPORT

The Profit Leak Diagnostic

What a $35.2M catalog's revenue report doesn't show you

84.5%
of profit comes from just 8 of 118 products

44
products earn far less than their revenue rank suggests

100%
of First Class orders arrive late

$643K
profit at risk if those 8 products lose 20% of their profit (modeled)

ABOUT THIS REPORT

This report applies the diagnostic to the public DataCo Supply Chain dataset: 180,519 order lines across three years. It shows the method and the kind of findings it produces. The findings describe that catalog, not your store.

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THE SHORT VERSION

Five findings, and what to do about each

Revenue is not the problem. Treating revenue as the measure of health is. Each page that follows shows the evidence for one finding, why it matters, and what to check first.

#	What the data shows	Evidence	First move
1	**Revenue rank and profit rank disagree.** 44 of 118 products earn far less than their revenue suggests. The SOLE E35 Elliptical is 36th by revenue and last by profit.	Observed	Rank products by profit, not revenue, in your weekly report.
2	**Discounting is the baseline.** The median product sells 94.5% of its orders with a discount, and discount depth shows no significant link to margin (p = 0.73).	Observed	Study the orders that still lose money after the discount.
3	**Profit rests on 8 products (84.5%).** A 20% fall in their profit would cost about $643K, 17% of the total.	Observed + Modeled	Lock supplier terms and protect fulfillment on those 8.
4	**First Class is late on 100% of orders,** by 1.00 day on average. The cheapest tier, Standard, is the most reliable.	Observed + Hypothesis	Check whether the promise or the process is wrong.
5	**28 low performers add 2.2% of revenue** and about $67K of profit, spread across 23 categories.	Observed	Sort them by cause before cutting any.

How to read the labels

OBSERVED MODELED HYPOTHESIS

Observed findings are counted directly from the data. **Modeled** findings are scenarios: they depend on the assumptions stated beside them and are not forecasts. **Hypothesis** marks a plausible explanation that the data alone cannot confirm.

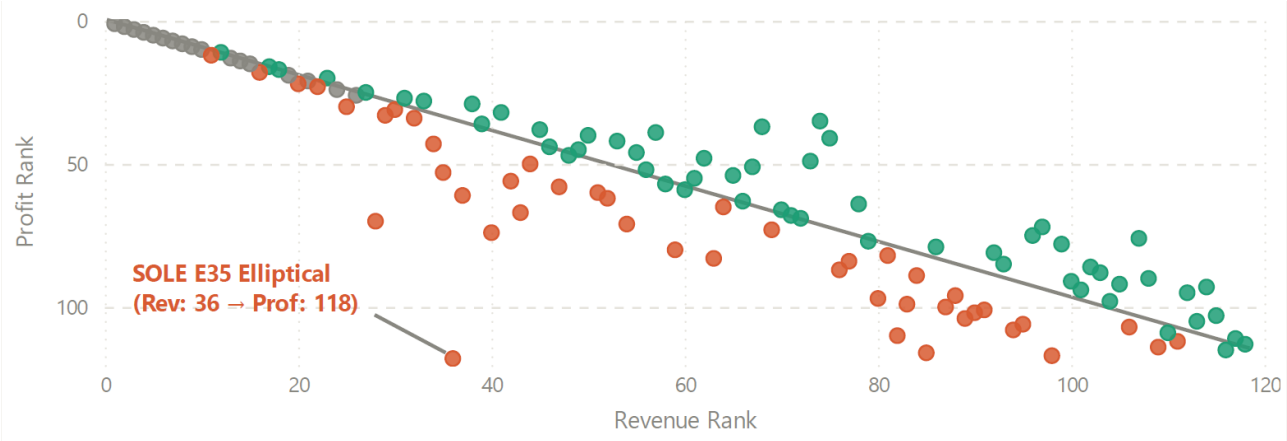
FINDING 1 OBSERVED

Revenue ranks products one way. Profit ranks them another.

36th
SOLE E35 Elliptical: rank by revenue

118th
its rank by profit, last of 118

44
products rank well below their revenue position on profit



Each dot is one product (118 in total), ranked by revenue (left to right) and by profit (top to bottom). Orange dots below the line sell better than they earn.

What the data shows

- **44 of 118 products** rank well below their revenue position on profit (the orange dots below the line).
- The **SOLE E35 Elliptical** and the **SOLE E25 Elliptical** are net negative across the full three years. A standard sales report would show the E35 as a respectable mid-tier seller.
- Every product in the catalog has recorded at least one loss-making order, including the best sellers.

WHY IT MATTERS

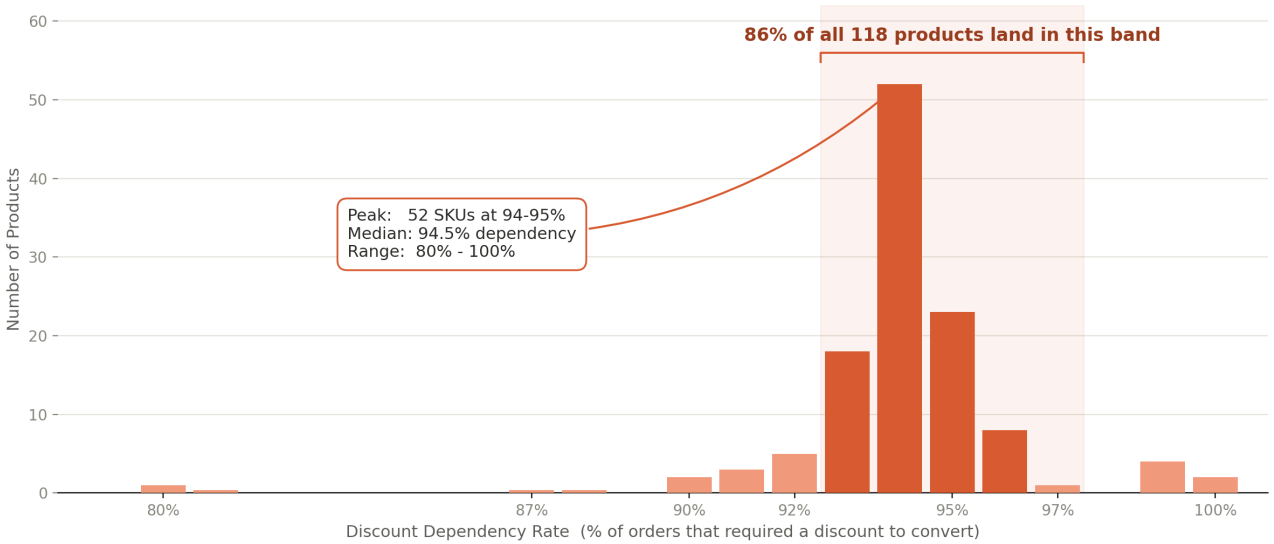
A revenue leaderboard rewards volume. If stock, ad spend and promotions follow that list, money keeps flowing to products that sell without earning.

WHAT TO CHECK FIRST

Add a profit-rank column next to revenue rank. Look first at products in your top 40 by revenue that sit in the bottom third by profit.

FINDING 2 OBSERVED

Discounting is the baseline, not a lever



Share of each product's orders that needed a discount to convert. 118 products, three years of orders.

What the data shows

- The median product sells **94.5%** of its orders with a discount. 86% of products fall between roughly 93% and 97%. The full range is 80% to 100%.
- Statistical testing found **no significant relationship between discount depth and margin** (p = 0.73). Discount depth does not explain which products earn less.
- **Limit:** this is a product-level test. It shows discount depth does not separate strong products from weak ones. It does not show that discounting is harmless.
- The four highest-revenue products are the most profitable in dollars, yet **94.4%** of their orders carry a discount and **18.5%** of their orders lose money.

WHY IT MATTERS

When almost every order is discounted, the discount is part of the real price. Margin read from list prices overstates what the business earns, and trimming promotions is unlikely to be a simple fix.

WHAT TO CHECK FIRST

List the orders that lose money even after the discount and find the price, cost and shipping combinations behind them. Then check what share of your own orders use a code.

FINDING 3 OBSERVED MODELED

Eight products carry 84.5% of profit

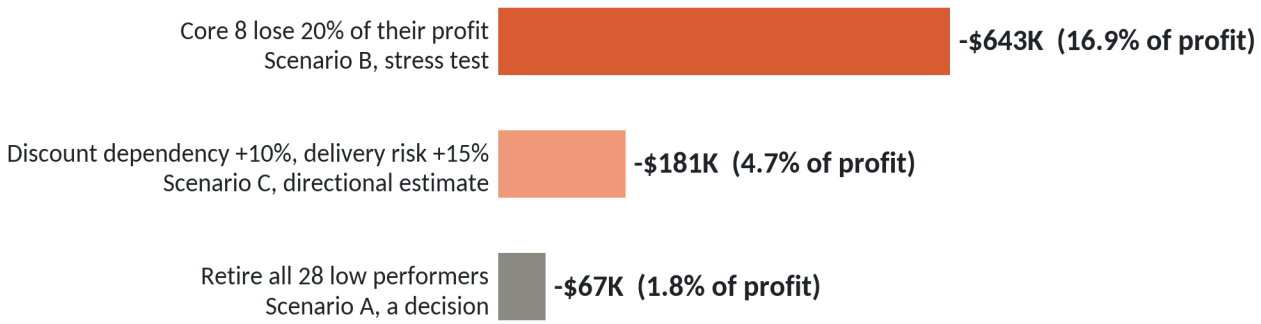


Grey: the other 110 products

Share of profit and revenue from the 8 core products versus the other 110.

What the data shows

- **8 of 118 products** generate 84.5% of profit and 84.7% of revenue. The other 110 together produce about $590K of the $3.81M baseline profit.
- **Stress test:** a 20% margin shock to those 8 products (a 20% fall in the profit they earn), from a supplier price rise or a logistics cost for example, removes about **$643K**, 16.9% of baseline profit. That is more than the other 110 products earn combined.
- A third scenario applies **+10% discount dependency and +15% delivery risk** and costs about $181K (4.7%). It is a directional estimate.



Profit lost against the $3.81M baseline. Scenario B applies a 20% margin shock to the 8 core products. Scenario C applies +10% discount dependency and +15% delivery risk. Both are illustrative stress tests, not forecasts. Scenario A is the decision to retire the 28 low performers.

WHY IT MATTERS

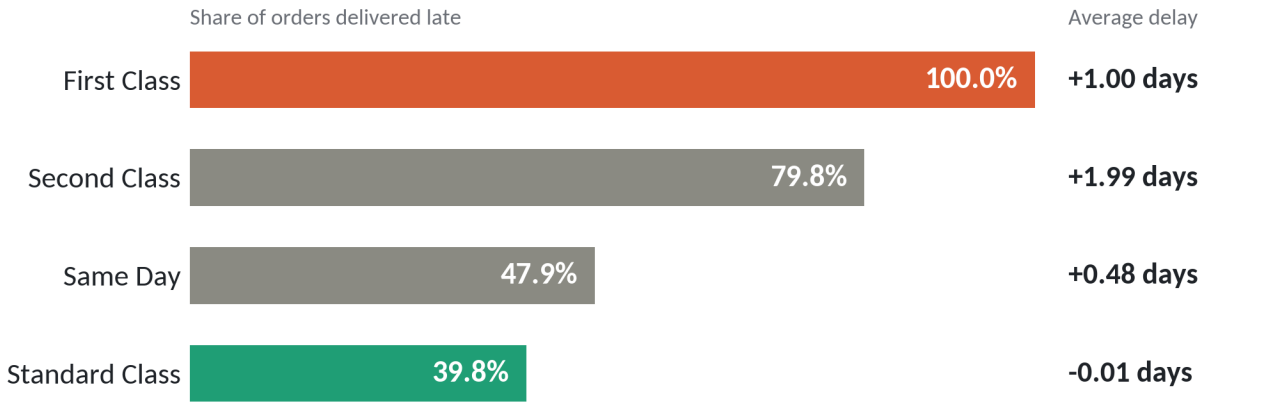
A single point of failure hides inside a 118-product catalog. A revenue report shows 118 healthy lines. The profit view shows one fragile core.

WHAT TO CHECK FIRST

Find the share of profit your top 5 to 10 products earn, then ask what happens to total profit if their margin drops 20%.

FINDING 4 OBSERVED HYPOTHESIS

First Class is late on every order, by about a day



Share of orders delivered later than scheduled, by shipping method. Average delay is in days; a negative number means early.

What the data shows

- **First Class:** 100.0% of orders late, 1.00 day on average. **Second Class:** 79.8% late with the longest average delay, 1.99 days. **Same Day:** 47.9%, 0.48 days. **Standard Class:** 39.8%, about on time.
- Standard Class, the cheapest option, is the most reliable method in the catalog.
- A rate of exactly 100% with an average delay of exactly 1.00 day is unusually clean. That pattern points to a fixed gap between the promised and the actual delivery time more than to day-to-day operational noise. **(Hypothesis:** confirm by looking at the distribution of delay days.**)**

WHY IT MATTERS

Customers paying a premium for speed are the ones most consistently disappointed. Late premium orders are the likeliest source of refund requests and poor reviews, although this dataset contains neither.

WHAT TO CHECK FIRST

Either fix the process so First Class meets its promise, or change the promise to match what is delivered. The delay distribution shows which problem you have.

FINDING 5 OBSERVED

28 products deserve a second look, not an automatic cut

28
products flagged (24% of the catalog)

2.2%
of total revenue

$67K
of profit they still contribute (1.8%)

23
categories affected (46% of all)

What the data shows

- These 28 products combine weak margins with high delivery failure. Together they generate **2.2% of revenue** and about **$67K (1.8%) of profit**.
- They sit across 23 categories, nearly half the taxonomy. This is not one bad category or one bad supplier.
- In aggregate they are not losing money. Retiring all 28 would give up about $67K of profit, before any saving in storage, handling or support. This dataset has no cost data for those, so those savings are not measured here.

Before cutting any of them, sort them by cause

If the cause is...	Then...
New, ramping up or seasonal	Keep it and review after one full sales cycle.
Weak margin from cost or discounting	Renegotiate the cost or reprice.
Held back by delivery failures	Fix fulfillment, then measure again.
None of the above, after checking	Retire it. The downside is small.

WHY IT MATTERS

A cut list without causes is hard to defend and can remove products that are new, seasonal or only held back by shipping. The downside is capped near $67K, which makes it safe to check first.

WHAT TO CHECK FIRST

Pull launch dates, monthly sales and supplier terms for each of the 28. Most will sort quickly into one of the four rows above.

WHAT TO DO NEXT

Four moves, in order of exposure

#	Move	What it involves	Why this order
1	**Protect the core 8**	Lock supplier terms, prioritize their fulfillment, and change their pricing only with evidence.	Largest exposure: about $643K in the stress test.
2	**Resolve First Class**	Fix the process behind it, or reset the promise to match what is delivered.	Customer-facing, and every order on that tier is affected.
3	**Triage the 28**	Sort them by cause with the four-row test, then retire only the confirmed dead weight.	Downside capped near $67K. Savings are unmeasured.
4	**Rank by profit**	Put profit rank beside revenue rank in every product report.	Cheap, and it stops the other three problems from hiding.

CHECK YOUR OWN STORE

Five checks you can run this week

You do not need this report's data to start. An order export and your product costs are enough.

- [ ] **Rank by profit.** Sort your products by profit instead of revenue. How many top-revenue products land in the bottom third?
- [ ] **Count the discounts.** What share of orders use a code, and what share of those orders still lose money?
- [ ] **Find the core.** What share of profit comes from your top 5 to 10 products? What happens if their margin drops 20%?
- [ ] **Test the promise.** For each shipping tier, what share arrives later than promised, and by how many days?
- [ ] **Diagnose before cutting.** For your lowest-profit products: are they new, seasonal, costly to deliver, or just weak?

METHOD, DEFINITIONS AND LIMITS

How this was built, and where it stops

180,519 order lines were loaded into a PostgreSQL warehouse and checked for data quality at each stage. Every figure was first established in SQL, then reproduced in Python and tested statistically. Because revenue and profit are heavily right-skewed, the tests are rank-based (Spearman, Kruskal-Wallis, Mann-Whitney). Findings count as confirmed only where tests passed at the 95% confidence level ($p < 0.05$). Where a relationship did not hold, it is reported as such.

Definitions

Low-performing ("zombie") product	The "Cut Candidates" quadrant: weak margin and high delivery failure relative to catalog medians. 28 of 118 products. Details in the project's decision log.
Core products	The 8 products that generate 84.5% of profit.
Loss-making order	An order line with negative profit.
Discount dependency	The share of a product's orders that used a discount.
Late delivery	An order that took longer than its scheduled shipping time. Average delay is actual minus scheduled days.
Revenue-trap product	A product whose profit rank sits well below its revenue rank, found by rank-divergence analysis (44 of 118).

Limits

- **One public dataset.** It illustrates the method. Your store will differ.
- **It shows what happened, not why.** Explanations are labeled hypotheses until tested.
- **Scenarios are stress tests** with stated assumptions. They are not forecasts.
- **No cost data for handling, storage, support or returns,** so operational savings are not measured.
- Dashboard walkthrough: youtu.be/BtNOyrPLW7o. Full technical build (SQL, Python, decision log): github.com/zeeshan-akram-ds/ecommerce-revenue-illusion-audit.

RUN THIS ON YOUR STORE

Your margin is already in your transaction data.

Send me an order export from Shopify or your sales platform and I will check the same five things on your numbers: which products earn, which only sell, and where your profit is exposed. Start with a free 30-minute profit call.

Book the call: calendly.com/zeeshanakram/data-diagnostic
Email: zeeshanakram1704@gmail.com
LinkedIn: linkedin.com/in/zeeshan-akram-ds